(A) phylolm::three.point.compute ($\mathcal{O}(n)$ Postorder Pruning, Ho & Ané 2014)

Tier 3 (Root $\rho = 294$): $t_\rho = 0 \implies \text{denom}_\rho = 1, \quad \text{vec11}_\rho = 4.9464$

$$\hat{\mu}_{\text{phylolm}} = \frac{P_{1,\rho}}{\text{vec11}_\rho} = \mathbf{w}_{\text{GLS}}^\top \mathbf{x}, \qquad Q_{\text{internal}}(\mathbf{x}) = PP_\rho - \frac{P_{1,\rho}^2}{\text{vec11}_\rho}, \qquad \hat{\sigma}_{\text{REML}} = \sqrt{\frac{Q_{\text{internal}}(\mathbf{x})}{147}}$$

$$\hat{\sigma}_{\text{MLE}} = \sqrt{\frac{Q_{\text{internal}}(\mathbf{x})}{148}}, \qquad \widehat{\text{SE}}(\hat{\mu}_{\text{phylolm}}) = \frac{\hat{\sigma}_{\text{REML}}}{\sqrt{\text{vec11}_\rho}} = 0.4496\,\hat{\sigma}_{\text{REML}}, \qquad \ln\det(\mathbf{S}) = \text{logd}_\rho = -112.33$$

$\mathbf{r}_\rho$ $(u = 294)$

$\Delta\tilde{t}_e(\boldsymbol{\theta})$

Tier 2 (Internal Nodes): $u \in V_{\text{int}} = \{148, \ldots, 294\}, \quad t_u = \Delta t_u, \quad \text{ch}(u) = \{l, r\}$

(2c) Rank-1 Schur-Complement Update ($\mathbf{r}_u$):

$$u_{1,u} = \frac{Y_{1,\text{ch}}}{\text{denom}_u}, \qquad u_{X,u} = \frac{Y_{X,\text{ch}}}{\text{denom}_u}, \qquad \text{vec11}_u = P_{\text{ch}} - \frac{t_u Y_{1,\text{ch}}^2}{\text{denom}_u}$$

$$P_{1,u} = P_{1,\text{ch}} - \frac{t_u Y_{1,\text{ch}} Y_{X,\text{ch}}}{\text{denom}_u}, \qquad PP_u = PP_{\text{ch}} - \frac{t_u Y_{X,\text{ch}}^2}{\text{denom}_u}$$

(2a) Child Accumulator ($\oplus$):

$$P_{\text{ch}} = \text{vec11}_l + \text{vec11}_r, \quad \text{logd}_{\text{ch}} = \text{logd}_l + \text{logd}_r$$
$$Y_{1,\text{ch}} = u_{1,l} + u_{1,r}, \quad Y_{X,\text{ch}} = u_{X,l} + u_{X,r}$$
$$P_{1,\text{ch}} = P_{1,l} + P_{1,r}, \quad PP_{\text{ch}} = PP_l + PP_r$$

(2b) Branch Precision Factor (denom_u):

$$\mathbf{denom}_u = 1 + t_u P_{\text{ch}} = 1 + t_u(\text{vec11}_l + \text{vec11}_r)$$
$$\text{logd}_u = \text{logd}_{\text{ch}} + \ln(\text{denom}_u)$$

Post-order $\mathcal{O}(n)$ $v = 0 \uparrow 294$

$\mathbf{r}_{l(u)}$ $\mathbf{r}_{r(u)}$

Tier 1 (Leaves $v \in V_{\text{leaf}}$): $t_v = \Delta t_v, \quad \mathcal{O}(1)$ **Init**

$l(u)$ $(t_l = \Delta t_{l(u)})$:

$$\text{logd}_l = \ln t_l, \quad u_{1,l} = \frac{1}{t_l}, \quad u_{X,l} = \frac{x_l}{t_l}$$
$$\text{vec11}_l = \frac{1}{t_l}, \quad P_{1,l} = \frac{x_l}{t_l}, \quad PP_l = \frac{x_l^2}{t_l}$$

$r(u)$ $(t_r = \Delta t_{r(u)})$:

$$\text{logd}_r = \ln t_r, \quad u_{1,r} = \frac{1}{t_r}, \quad u_{X,r} = \frac{x_r}{t_r}$$
$$\text{vec11}_r = \frac{1}{t_r}, \quad P_{1,r} = \frac{x_r}{t_r}, \quad PP_r = \frac{x_r^2}{t_r}$$

(B) $\Delta t_e \mapsto \Delta\tilde{t}_e(\boldsymbol{\theta})$ & TreeMPNN

1. Branch Transform $\mathcal{T} \mapsto \tilde{\mathcal{T}}(\boldsymbol{\theta})$:

- OU(α): $\tilde{T}_u(\alpha) = \frac{1-e^{-2\alpha T}}{2\alpha} e^{-2\alpha(T-t_u)}$
- EB(r): $\Delta\tilde{t}_e(r) = \frac{e^{r t_{\text{dst}}} - e^{r t_{\text{src}}}}{r}$
- Pagel $(\lambda, \kappa, \delta)$: $\mathbf{V}(\boldsymbol{\theta}) \mapsto \Delta\tilde{t}_e(\boldsymbol{\theta})$

2. $\mathcal{O}(n)$ Pruning vs. $\mathcal{O}(n^3)$ Cholesky:

- Cholesky $\mathbf{V}(\boldsymbol{\theta}) = \mathbf{L}\mathbf{L}^\top$: $\mathcal{O}(n^3)$ time, $\mathcal{O}(n^2)$ mem
- phylolm 3-pt: $\mathcal{O}(n)$ time, $\mathcal{O}(n)$ mem
- $\max_k \left|\hat{\mu}_{\text{phylolm}} - \hat{\mu}_{\text{Cholesky}}\right| = \mathbf{3.55 \times 10^{-15}}$

3. Isomorphism phylolm $\leftrightarrow$ TreeMPNN:

$$\text{vec11}_u^{-1} \longleftrightarrow v_u = \frac{d_L(u) d_R(u)}{d_L(u) + d_R(u)}$$
$$\frac{P_{1,u}}{\text{vec11}_u} \longleftrightarrow H_u = \frac{d_R(u) H_{l(u)} + d_L(u) H_{r(u)}}{d_L(u) + d_R(u)}$$
$$C_u = \frac{H_{l(u)} - H_{r(u)}}{\sqrt{d_L(u) + d_R(u)}}, \qquad d_c(u) = (v_c + \Delta t_c) e^{g_{c,u}}$$